FACULTY OF COMPUTER SCIENCE AND MATHEMATICS BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOURS

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

PROJECT TITLE:

GROUP 7

CLOUD-NATIVE LEARNING MANAGEMENT SYSTEM (LMS)

(CLOUD-NATIVE/ LAYERED ARCHITECTURE)

INDIVIDUAL LOGBOOK

PREPARED FOR:

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PREPARED BY:

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1.0 Team Information

Project Title: Cloud-Native Learning Management System (LMS)

Group Number: 7 (K2)

Team Members:

Name	Role
Nur Alia Farisya Binti Mohd Khairy	System Architect & DevOps Lead
Nur Aliya Farhani Binti Mohd Nasri	Backend & Database Lead
Zihan Farisya Binti Mohd Zaini	Frontend & UI Developer

2.0 Weekly Progress

2.1 Week 4

Item	Description
Week	Week 4 (5 April - 9 April 2026)
Weekly Activities	- Participated in initial team discussions to define the project scope for the LMS - Began drafting the introduction section for deliverable 1, outlining the system's purpose and architecture.
Individual Contribution	- Authored the "Introduction" section of the project proposal, framing the LMS as a web-based application with a layered architecture and cloud-native deployment on Aiven.io.
Challenges	- Ensuring the introduction clearly and concisely described the layered architecture concept without being too technical.
Action Plans	- Review the proposal draft to ensure all sections are cohesive and align with the proposed architectural style.

2.2 Week 5

Item	Description
Week	Week 5 (12 April - 16 April 2026)
Weekly Activities	- Finalized and polished the deliverable 1: Project proposal - Successfully submitted the proposal. - Began researching UI frameworks and technologies for the presentation layer.
Individual Contribution	- Finalized the "Introduction" and "Team Member Roles" sections. - Proposed the use of HTML5, CSS3, and JavaScript for the presentation layer, justifying their role in building the static and interactive UI components.
Challenges	- None significant; the team was well-organized and met the submission deadline.
Action Plans	- Transition to Phase 2 (Architecture Design) and begin planning the UI structure for the registration and login pages.

2.3 Week 6

Item	Description
Week	Week 6 (19 April - 23 April 2026)
Weekly Activities	- Worked with the team on Phase 2: Architecture design. - Focused on identifying the components for the presentation layer and defining their responsibilities.
Individual Contribution	- Mapped out the specific UI components: the user authentication interface (register.jsp, login.jsp) and the course dashboard UI (dashboard.jsp). - Defined the data flow for these UI components.
Challenges	- Understanding the data structures required from the backend to effectively design the dynamic dashboard interface.
Action Plans	- Collaborate with the backend lead to finalize the data payloads for user registration and course retrieval.

2.4 Week 7

Item	Description
Week	Week 7 (26 April - 30 April 2026)
Weekly Activities	- Finalized the architecture diagrams for deliverable 2. - Consolidated all items into the final deliverable 2: Architecture design document. |
Individual Contribution	- Designed the system context diagram, clearly illustrating the user roles (Learners, Educators, Administrators) interacting with the LMS system. - Assisted in drafting the component/service architecture diagram to show the UI components connecting to the application and data access layers.
Challenges	- Ensuring all four required diagrams (system context, component, interaction, deployment) were visually consistent and accurately reflected the architecture.
Action Plans	- Submit deliverable 2 on time and prepare for Phase 3 (Prototype implementation).

2.5 Week 8

Item	Description
Week	Week 8 (10 May - 14 May 2026)
Weekly Activities	- Began the implementation phase by setting up the frontend development environment. - Started coding the static structure of the user registration and login pages using HTML5 and CSS3.
Individual Contribution	- Wrote the initial `.jsp` layouts for the registration and login interfaces, as shown in Figures 1-6 of the final report. - Implemented a responsive design and integrated HTML5 native input constraints for client-side validation.
Challenges	- Ensuring the client-side validation (FR-01, FR-02) provided a good user experience without interfering with backend server-side validation.
Action Plans	- Complete the base UI for all pages and begin integrating them with the backend Java Servlets.

2.6 Week 9

Item	Description
Week	Week 9 (17 May - 21 May 2026)
Weekly Activities	- Integrated the frontend JSP pages with the backend logic developed by the Backend Lead. - Conducted internal codebase architecture alignment checks.
Individual Contribution	- Mapped form actions to the correct servlet endpoints (e.g., `register.jsp` submits to `AuthServlet`). - Ensured the UI elements correctly passed the required parameters (username, email, password) to the backend.
Challenges	- Coordinating variable naming conventions between the frontend form inputs and the backend request parsers to ensure data integrity.
Action Plans	- Begin work on the dynamic course dashboard UI to display data fetched from the Aiven.io cloud database.

2.7 Week 10

Item	Description
Week	Week 10 (24 May - 28 May 2026)
Weekly Activities	- Developed the course dashboard UI as shown in Figure 7 of the final report. - Integrated session-data to display personalized information on the dashboard.
Individual Contribution	- Built the dynamic JSP page that displays the user's name (from the session) and a list of courses. - Ensured that the UI is protected and only accessible when a user has an active session, thus implementing FR-05.
Challenges	- Designing the dashboard to be clear and intuitive while handling potential edge cases, such as a user with no enrolled courses.
Action Plans	- Test the UI's interaction with the live Aiven.io database and refine the user experience.

2.8 Week 11

Item	Description
Week	Week 11 (31 May - 4 June 2026)
Weekly Activities	- Prototype completion and submission. - Consolidated the NetBeans project source directory.
Individual Contribution	- Finalized all UI views (Login, Registration, Dashboard). - Verified that the UI flow correctly implements the session logic (redirecting unauthenticated users to the login page). - Compiled submission evidence (screenshots) for deliverable 3.
Challenges	- Managing edge-case execution paths where a corrupted string input could potentially disrupt the background screening worker threads.
Action Plans	- Transition into phase 5 (Evaluation, Code Refinement and Final Report Writing).

2.9 Week 12

Item	Description
Week	Week 12 (7 June - 11 June 2026)
Weekly Activities	- Conducted comprehensive internal testing and debugging on the core integrated packages. - Began writing the final project report.
Individual Contribution	- Authored the "Architecture Evaluation" section for the final report. - Evaluated the prototype against the non-functional requirements (Maintainability, Scalability, Availability, Security, Portability, Modularity). - Documented the system's adherence to the layered architecture and its strengths/weaknesses.
Challenges	- Standardizing the format and language used across all sections of the final report.
Action Plans	- Compile the final project report chapters and assemble individual reflection statements.

2.10 Week 13

Item	Description
Week	Week 13 (14 June - 18 June 2026)
Weekly Activities	- Completed Deliverable 4: Project Report & Individual Logbook. - Finalized the comprehensive report and individual contributions.
Individual Contribution	- Finalized the "Architecture Evaluation" section. - Compiled and formatted this individual logbook. - Participated in team discussions to finalize the "Conclusion and Reflection" section.
Challenges	- Restructuring and condensing extensive code segments into clear, concise code-snippet figures for the formal text.
Action Plans	- Submit deliverable 4 and coordinate with team members to run rehearsal runs of our system for the final demonstration.

3.0 GenAI Usage Record

Field	Details
Date (Week)	Week 4 (5 April - 9 April 2026)
Tool Used	Google Gemini
Purpose	To help draft a clear and structured problem statement for a traditional LMS.
Prompt Used	"*Help me draft a problem description for a traditional university Learning Management System, highlighting issues with infrastructure bottlenecks, session management, and maintenance overhead*."
Output Summary	Provided a structured list of key problem areas: Infrastructure bottlenecks, lack of dynamic session management, and data vulnerability, with detailed explanations.
How It Was Used	Used as a foundation to construct the initial "Problem Description" section of our proposal. I rewrote and elaborated on the points to fit the specific context of our project.

Field	Details
Date (Week)	Week 6 (19 April - 23 April 2026)
Tool Used	Google Gemini
Purpose	To assist in drafting comprehensive and well-structured functional requirements.
Prompt Used	"*Based on user authentication and course dashboard modules, generate a list of functional requirements for an LMS*."
Output Summary	Provided a list of 8 functional requirements that were generic in nature but served as a good starting point.
How It Was Used	I used this list as a template and edited the items to be specific to our prototype's capabilities, resulting in FR-01 to FR-08.

Field	Details
Date (Week)	Week 8 (10 May - 14 May 2026)
Tool Used	Google Gemini
Purpose	To explore best practices for structuring JSP pages for a clean and maintainable user interface.

Prompt Used	"*What are the best practices for designing a clean and responsive login and registration page using HTML5, CSS3, and JSP*?"
Output Summary	Provided recommendations on using modern HTML5 input types for validation, CSS frameworks for responsive design, and structuring JSP files to separate logic from presentation.
How It Was Used	Guided the development of the registration (`register.jsp`) and login (`login.jsp`) pages, ensuring they were responsive and utilized native HTML5 validation constraints.

Field	**Details**
Date (Week)	Week 12 (7 June - 11 June 2026)
Tool Used	Google Gemini
Purpose	To help structure the Architecture Evaluation section and find appropriate metrics for analysis.
Prompt Used	"*How do I evaluate a layered web application architecture for maintainability, scalability, and security in an academic report*?"
Output Summary	Provided a framework for evaluation, suggesting criteria such as separation of concerns, single points of failure, and use of security best practices like parameterized queries.
How It Was Used	Formed the basis of the "Architecture Evaluation" section in our final report. I used the suggested criteria to assess our own system and created a structured table for the findings.

Field	**Details**
Date (Week)	Week 13 (14 June - 18 June 2026)
Tool Used	Google Gemini
Purpose	To check for consistency and clarity in my writing for the final report.
Prompt Used	"*Proofread this paragraph on the architecture evaluation for clarity and grammatical errors*."
Output Summary	Provided a polished version of the text with minor grammar and syntax corrections.
How It Was Used	Used to refine the final draft of my sections in the "Architecture Evaluation" and "Conclusion"